WORKED SOLUTION

STEP III 1999, Question 1

Pure • Roots of Cubic Equations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q1

The key idea

Everything here comes from Vieta's formulas: for the cubic $x^3 - px^2 + qx - r = 0$ with roots α, β, γ,

$$\alpha + \beta + \gamma = p, \qquad \alpha\beta + \beta\gamma + \gamma\alpha = q, \qquad \alpha\beta\gamma = r.$$

A geometric progression is centred on its middle term *multiplicatively*, so the product r and the ratio q/p pick out the middle root. An arithmetic progression is centred *additively*, so the sum p picks out the middle root instead.

Solution

Part (i)

Suppose the roots are a/k, a and ak. For a/k to make sense we need $k \neq 0$, and since the product of the roots is $r \neq 0$, we also have $a \neq 0$. Vieta's formulas give

$$\begin{aligned} p &= \frac{a}{k} + a + ak = a\left(\frac{1}{k} + 1 + k\right), \\ q &= \frac{a}{k} \cdot a + a \cdot ak + \frac{a}{k} \cdot ak = a^2\left(\frac{1}{k} + 1 + k\right), \\ r &= \frac{a}{k} \cdot a \cdot ak = a^3. \end{aligned}$$

The same bracket appears in p and q. It is nonzero because $p \neq 0$, so we can divide:

$$\frac{q}{p} = a.$$

So q/p is a root (the middle one). Then $q = ap$, and so

$$q^3 = a^3p^3 = rp^3, \qquad \text{that is,} \qquad q^3 - rp^3 = 0,$$

as required.

Part (ii)

Now suppose $r = q^3/p^3$, and write $a = q/p$, so that $r = a^3$. Substituting $x = a$ into the cubic, and using $p\,a^2 = q^2/p = q\,a$,

$$a^3 - pa^2 + qa - r = a^3 - qa + qa - a^3 = 0,$$

so q/p is a root.

Since $r \neq 0$, we have $q \neq 0$, so $a \neq 0$. Call the other two roots β and γ. The product of all three roots is r, so

$$a\beta\gamma = r = a^3 \quad \Longrightarrow \quad \beta\gamma = a^2 = \left(\frac{q}{p}\right)^2,$$

dividing by $a \neq 0$.

? **Ask yourself** How do we turn $\beta\gamma = a^2$ into the form a/k, a, ak?

We need to choose k. Since $\beta\gamma = a^2 \neq 0$, neither β nor γ is zero. Define $k = \gamma/a$, so that $\gamma = ak$ and $k \neq 0$. Then

$$\beta = \frac{a^2}{\gamma} = \frac{a^2}{ak} = \frac{a}{k}.$$

So the roots are a/k, a and ak, each obtained from the one before by multiplying by k. They are in geometric progression. (The ratio k may be complex when β and γ are not real. The argument does not need them to be real.)

Together, parts (i) and (ii) show that, when $p \neq 0$ and $r \neq 0$, the roots are in geometric progression if and only if $q^3 = rp^3$.

Part (iii)

Necessity. Suppose the roots are $b - d$, b and $b + d$. Their sum is $3b = p$, so the middle root is $b = p/3$. Substituting $x = p/3$ into the cubic and multiplying by 27:

$$p^3 - 3p^3 + 9pq - 27r = 0 \quad \Longleftrightarrow \quad 2p^3 - 9pq + 27r = 0.$$

Sufficiency. Conversely, suppose $2p^3 - 9pq + 27r = 0$. Reversing the step above, $p/3$ is a root. The other two roots then have sum $p - p/3 = 2p/3$, so we can write them as $p/3 - d$ and $p/3 + d$ for some (possibly complex) d. The three roots $p/3 - d$, $p/3$, $p/3 + d$ are in arithmetic progression.

Hence a necessary and sufficient condition is

$$\boxed{2p^3 - 9pq + 27r = 0}.$$

Checks

- Roots $1, 2, 4$ (a geometric progression) give $p = 7$, $q = 14$, $r = 8$. Then $q/p = 2$ is the middle root, and $q^3 = 2744 = 8 \cdot 343 = rp^3$.

- Roots $1, 2, 3$ (an arithmetic progression) give $p = 6$, $q = 11$, $r = 6$, and $2 \cdot 216 - 9 \cdot 66 + 27 \cdot 6 = 432 - 594 + 162 = 0$. The geometric progression $1, 2, 4$ gives $686 - 882 + 216 = 20 \neq 0$, as it should.
- We also tested all three parts numerically on thousands of random cubics, including ones with complex roots. All agreed to within 10^{-11}.

Common slips

- Dividing by $1/k + 1 + k$ without saying why it is nonzero. It is nonzero because it is p/a and $p \neq 0$.
- In part (ii), showing only that q/p is a root and that $\beta\gamma = (q/p)^2$, and then asserting a geometric progression without exhibiting the common ratio.
- In part (iii), proving only one direction. "Necessary and sufficient" needs both.

Beyond STEP

Vieta's formulas say that the coefficients of a polynomial are the *elementary symmetric functions* of its roots. A classical theorem says that every symmetric polynomial in the roots can be written in terms of these. This is the starting point of Galois theory, which explains when the roots of a polynomial can be found by radicals. Incidentally, the substitution $x = y + p/3$ removes the x^2 term from the cubic, and the constant term of the resulting depressed cubic is a multiple of $2p^3 - 9pq + 27r$. Allenby's *Rings, Fields and Groups* is a good place to see where these ideas lead.

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